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Mediating Irregularity: Towards a Design Method for Spatial Structures Utilizing Naturally Grown Forked Branches

Abstract. Computational methods enable the use of the complexity of material logics as generative parameter in an experimental form finding process for space defining structures. Examples are the fibrous structure and growing patterns of trees, which result in wood offering a vast spectrum of principles that have the potential to inform and define complex architectural configurations. Based on such assumptions, this paper presents an exploration of the relationship of material and space by utilizing naturally-grown forked branches as discrete y-shaped elements in a spatial aggregation with the aim of forming an architectural environment.

Keywords: 3D Scan, Naturally grown form, Complex Wood Structures, Spatial frameworks, discrete element aggregation, constraint based design..

1. Background

The inherent material properties of wood, such as grain direction, can be used as generative parameters for innovative timber structures utilizing the naturally optimized form of forked branches (Müller et al., 2014) (Mattheck, 1998) as structural nodes. Within architectural research, few attempts have been made to harness the potential for utilizing such potential (Schindler et al, 2013) (Von Buelow et al., 2018). One project in particular, the “Wood Chip Barn”, explored an approach where y-shaped wooden forks could be used to approximate a desired structural formation (Mollica and Self, 2016). This paper relies on a similar approach, but does not assume a pre-designed scheme, aiming rather at exploring the emergent formations embedded in the material logic. Those structures are not designed to a specific use at first, but are intended to suggest potential architectural applications based on their specific spatial and geometric properties. Geometry driven Architectural strategies similar to the “oblique function”(Parent and Virilio, 1966) could be developed resulting in novel architectural space concepts.

One key challenge in this project is working with a constrained inventory of discrete elements, each unique, yet similar. The complexity, irregularity and coincidence can be coped with by employing digital tools for means of control and a flexible design methodology. This challenge is understood as an advantage and potential for experimentation and novelty. Similar topics are explored within the field of Inventory-constrained design methods (Bukauskas et al., 2018) (Bruetting et al., 2018). However, beyond the obvious ecological advantage of an increased percentage in tree-part utilization, here this method is employed as a generative parameter in an experimental form finding process (Fig.1).



Fig. 1 spatial aggregation from a set of 3D-scanned hornbeam branch forks

1. Methods

With the aim to create unique, non-standard spatial structures at architectural scale,, the proposed workflow assumes a collection of physical branch parts and their distinct properties as starting point and design driver. This aims at exploring the design and performative potential of a specific set of unique elements and the relative emerging formations, rather than relying on a predetermined design, for which elements have to be adapted.

The design and manufacturing process is articulated in different steps:

1. Digitization: 3D scanning of real branches and automated features extraction from the scanned meshes.
2. Categorisation: Clustering of the scanned geometries according to geometric features, and generation of “proto-parts” as averages between clustered branches.
3. Discrete Element Aggregation: Generation of design structures using recursive aggregation of the generated proto-branches.
4. Population: Replacement of “proto-branches” with real branch geometries, according to specific design and performative parameters (i.a. according to length to gain denser (smaller) or open (higher) clusters/cells, or by cross sections according to structural analysis)
5. Relaxation: Compensation of structure gaps using a force-based relaxation of the whole model
6. Detailing for Fabrication: Original meshes placement and post-processing for fabrication

1.1. Geometry

With the aim of activating the structural potential embedded in the material logic of bifurcating strands of fibers, the branching parts are joined following the extension of the wood grain orientation. However, deviating from the internal logic of trees, which can be represented as an L-system with open ends, branches are assembled to form closed loops for improved structural integrity and general bracing (Allner and Kroehnert, 2018). The resulting cellular formations are intended to provide frameworks for architectural purposes. Considering the given principle of joining bifurcating elements, the most compact and thus stable cell form is a tetrahedron (Fig. 2). Out of various geometric aggregation principles that are being explored within the project, both regular and irregular, this paper focuses on one specific instance of a topologically repetitive/regular lattice of tetrahedron cells as a case study.

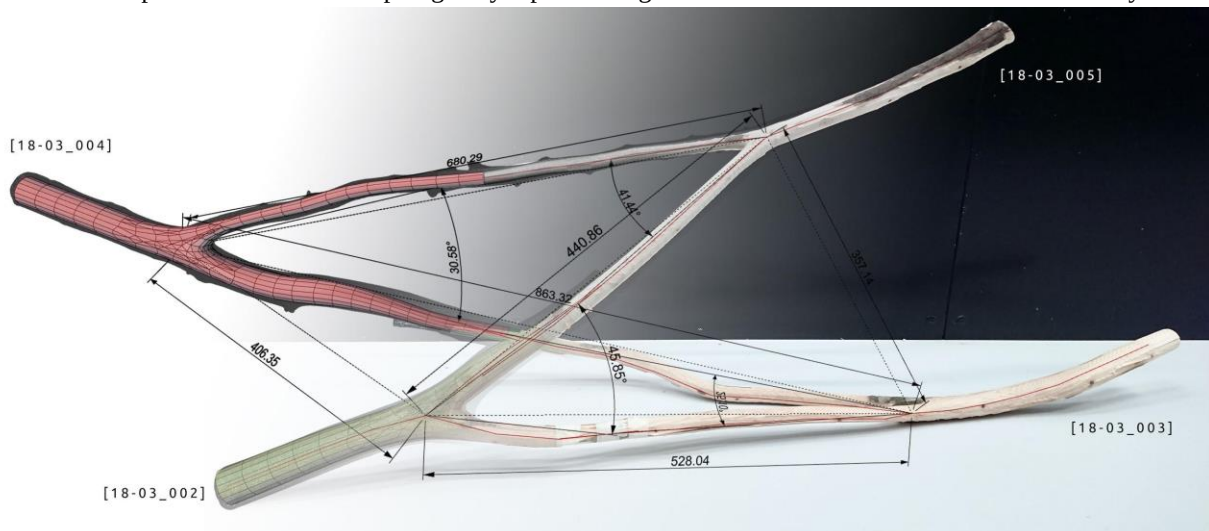


Fig. 2 tetrahedron assembly composed of four CNC-milled hornbeam branch forks

1.2. Scanning and Features Extraction

For easier/automated processing only Y-shaped branches were considered for this study. Scanning was done with the handheld Artec Eva structured light 3D scanner and meshed within the the Artec Studio Software. After flush-cutting the ends of the individual branch arms, they were prepared by physically marking 3 points on each flat end (Fig. 3) and picked up by the scanner as part of the texture.

In an automated process centerline or axis models are generated from the resulting mesh geometries by connecting the centroids of co-parallel section outlines with polylines (Fig.3). These serve as geometric reference for matching the digital and the analog models at any step of the process and allow to monitor the change of shape caused by in- or decrease in humidity to achieve high precision during future CAM processing despite the complex irregular form.

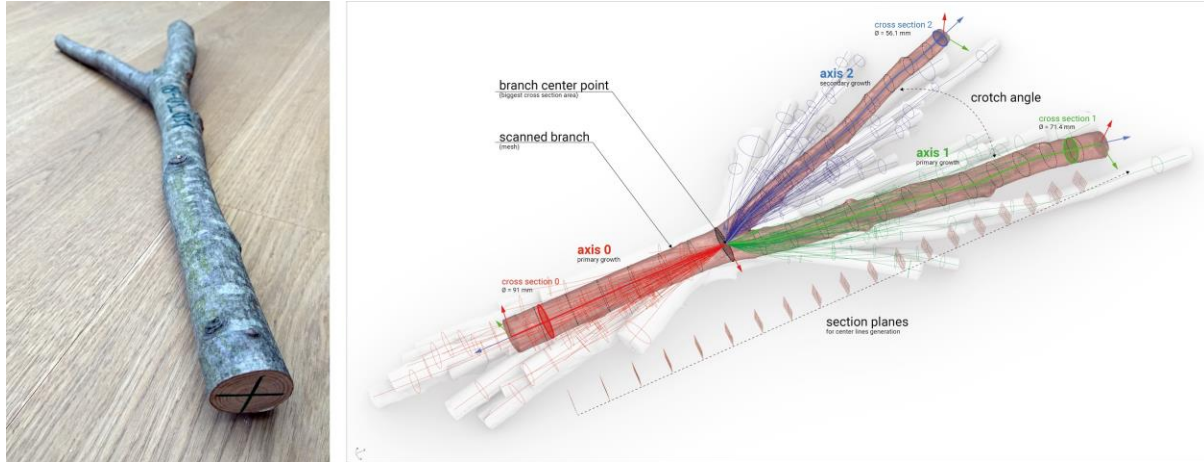


Fig. 3 marked branch for 3d scanning and digital data generated from scanned set

1.3. Catalog Clustering

The scanning and features extraction process returns a catalog containing a large number of unique elements, each with similar topology, but different dimensions and orientations of the individual branches. In order to reduce the variability of parts, which would lead to high computation times during structure aggregation, a clustering procedure is proposed to group the full catalog in sub-groups, according to geometric similarities. The process relies on a K-mean clustering algorithm (Vora and Oza, 2013), implemented in Python, which creates a number of sub-groups. The algorithm can be fed with different geometric data extracted from the branches. In a first instance, only dimension of each branch and its average crotch angle were used. In a further iteration, each part has been represented as a list of coordinates, obtained by subdividing the branches in points. This method allowed increased accuracy of the reconstructed results, preserving not only 2D information, but also the full 3-dimensional articulation of the branches (Fig.4).

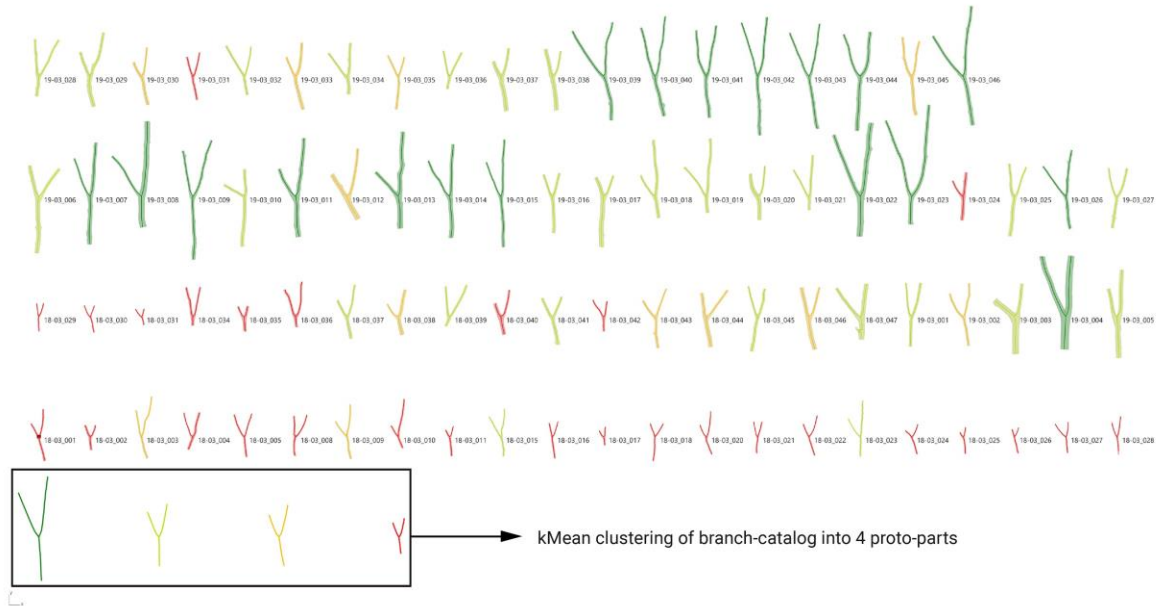


Fig. 4 *kMean clustered set of scanned branches into 4 groups represented by one proto-part each*

The algorithm can be tuned by changing the resolution of the provided points, as well as the clustering iterations. Besides the clustered geometries, the algorithm returns also, for each cluster, one “proto-part”, generated as the average of the geometric features of the parts in each cluster. These “proto-parts” are then used to generate the aggregated structure.

1.4. Structure Generation through Recursive Aggregation

In order to generate aggregated structures from discrete branch geometries, the Wasp plug-in for Grasshopper (Rossi and Tessmann, 2018) has been used. Through it, the “proto-parts” geometries have been converted in Wasp “Parts”, with specific connectivity topology specified. The tool allows then to iteratively aggregate the “proto-parts” into a structure, according to predefined connectivity rules. Different methods are available to control the outcome, including geometric constraints and scalar fields as aggregation drivers.

In order to apply Wasp within the context of the project, adaptations of the tool have been implemented. The first feature was the implementation of a novel part type, based on the axial polylines of each branch, rather than on their 3d-scanned mesh geometry. By implementing a novel collision solver, based on proximity between each branch polylines rather than on mesh clash, it was possible to highly improve the speed of the aggregation process, as well as account for the specificity of branch geometry. Additionally, a novel aggregation method, based on goals specific to the branch structurers, was also developed. The main feature of such method relies on minimizing the distance between the open-ends of a fork and the already aggregated branches, with the aim of creating closed structural loops within the aggregation. Finally, the possibility of using a part catalog, where each part can be used only a limited number of times, has been implemented. The part catalog was defined by assigning a number of parts for each proto-part, which corresponded to the number of original branches contained in the proto-part cluster. In this way, the final aggregated structure would contain a number of parts which would match the number of available branches, as well as their geometric features.

1.5. Design control: Proto-parts Replacement

The approximated formation is refined towards a final model by introducing the distinct elements from the available set. Each proto-part within the preliminary aggregation is replaced by the axis model of a real specimen from the inventory.

Various selection criteria are applicable to control the parametric population of the preliminary aggregation by unique elements. There are two domains of sorting, the location of each part as well as the features of each unique part in the catalog can be categorized. One instance of a sorting parameter/category within one domain is combined with a sorting parameter/category of the other in combination in order to produce meaningful results. An example would be the sorting of locations according to global z-value (location in height) combined with a sorting of parts by size, effecting an aggregation with larger parts at the bottom and smaller parts at the top.

1.6. Gap Compensation via Force-based Relaxation

Once the actual parts are positioned in the model the continuity of the preliminary proto-aggregation is compromised.



Fig. 5 aggregation (Wasp), replacement with scanned data and relaxation (Kangaroo) process

Geometrical differences between the unique elements and the relative proto-part, representing the respective geometric cluster, result in (an altered aggregation) with branch ends mismatching.

These local discontinuities are compensated by deforming the overall formation using a relaxation algorithm. The aggregation topology is translated into a constraint-based model in Kangaroo, a Live Physics engine plug-in for Grasshopper (Senatore and Piker, 2015). Such a model is created by converting each branch-part into a rigid body, connecting open ends with springs and defining a ruleset for geometric constraint solving. In an iterative simulation process the spring elements between parts compress, causing branch ends to overlap. Simultaneously the end segments of each two connected parts are constrained to align colinearly. This allows to compensate the gaps created by replacing the proto-parts with the original parts, as well as to create an overlap between the branches, where to create a connection joint (Fig. 5).

Finally, the 3D scanned mesh models are mapped to the locations of their respective axis model.

The resulting aggregated model is further refined and processed towards fabrication.

2. Joints

The overlap in the model allows joining parts without extension elements.

In this project only one type of joint is required, connecting (only) two elements in each location. Spatial complexity is introduced by the geometry of the part, which in return is joined in a simple fashion.

However when connecting parts different problems have to be addressed, structural integrity and stability, geometric

limitations/limited degrees of freedom when physically connecting parts and an assembly process (with as little additional support structure as possible).

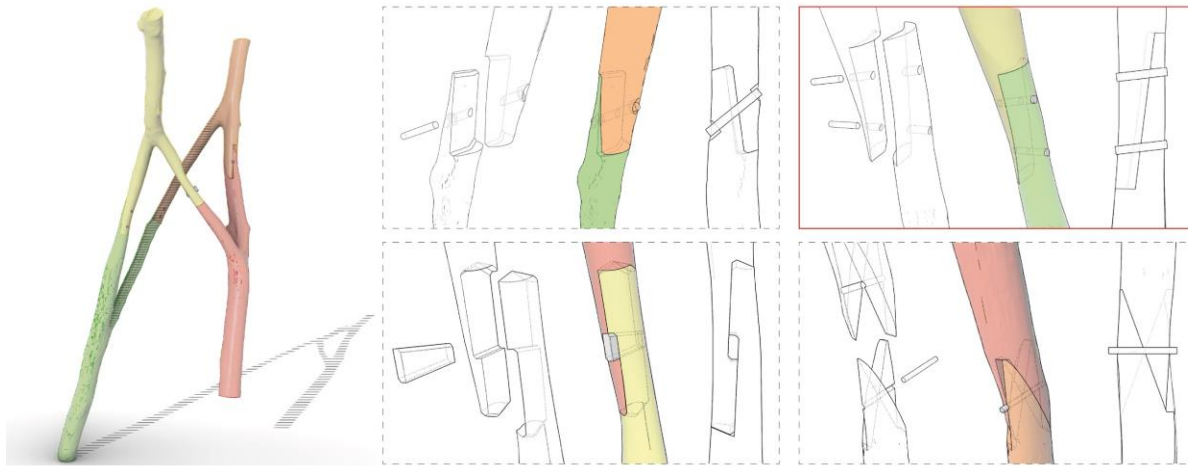


Fig. 6 joint study for the production of a prototype from 4 scanned branches

From various studies (Fig. 6) a half lap joint is chosen for a first demonstrator (Fig. 7). The joint employed here is a specific modification of a typical half lap extension joint. The lap surfaces have a conical cross section to improve stability in lateral direction as well as easing the assembly process by snapping into place. This type allows for sufficient freedom of movement for the assembly process whilst most other joints come with restricted/constrained assembly mechanics. For optimal traction the joints should be glued, for a flexible scenario detachable connections bolts could be used to achieve traction by compression and friction.

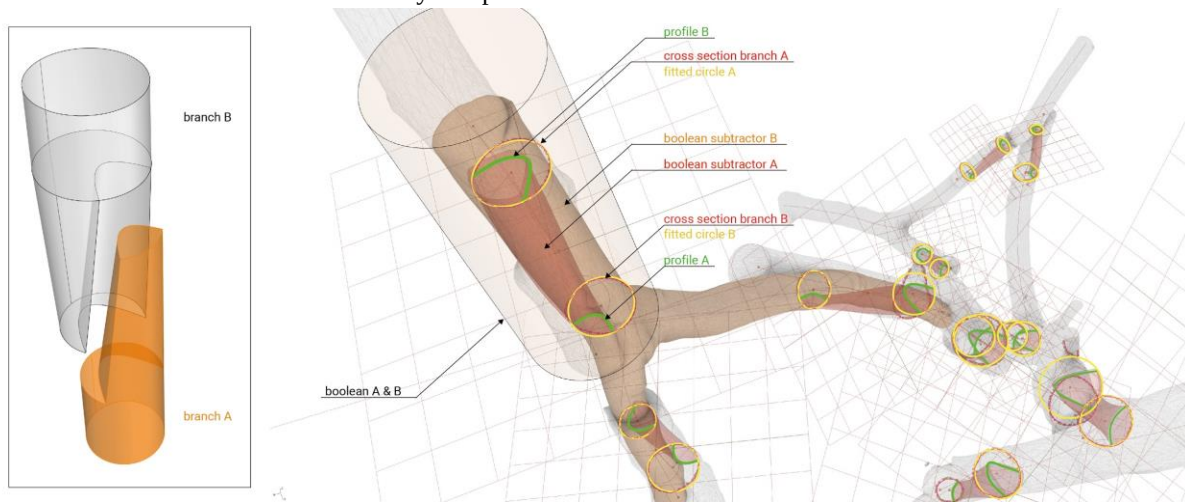


Fig. 7 automated generation of "curved" half-lap joints for the production of a demonstrator

3. Results

By using only one symmetrical proto-part composed of three line segments, representing an average of all branch elements in the set, regular cellular formations can be achieved. By applying a distinct ruleset (of connection angles) in Wasp open ends of aggregation strands regularly meet in one point, effectively closing the loop and forming a space cell.

From many possible cellular forms, an aggregation of tetrahedral cells is chosen to be further investigated (Fig. 8).

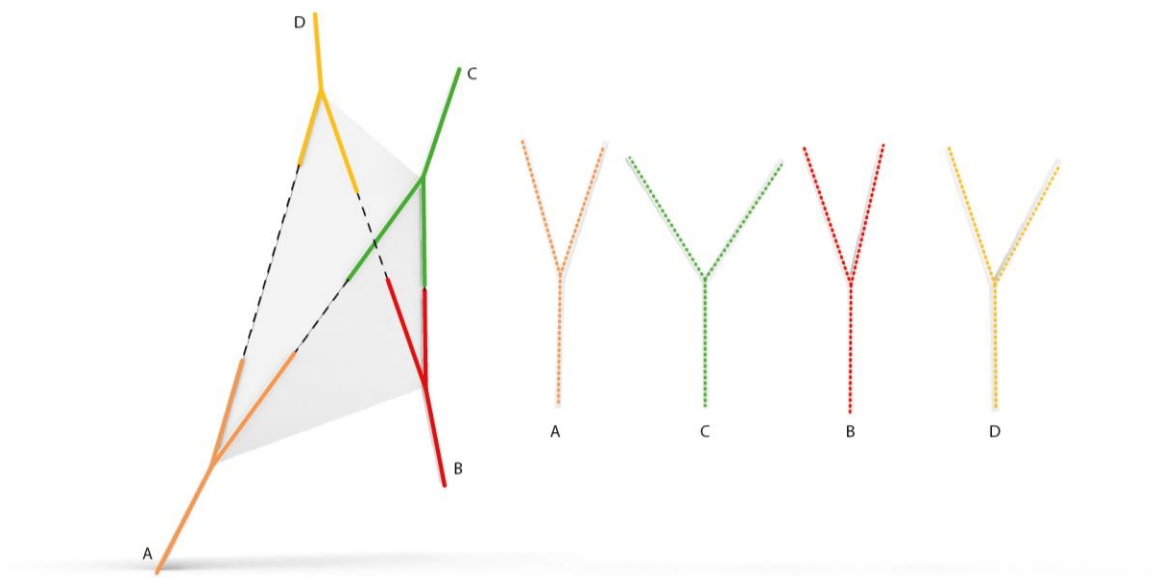


Fig. 8 tetrahedron from four Y-shaped branches

A tetrahedral form is the most compact spatial cell to be formed by y-shaped elements and thus performs as bracing link within a spatial framework.

the various sorting principles controlling the replacement of proto-part instances by members of the set can be freely applied across the entire formation, allowing for a high level of design control.

The predefined topology is a strong constraint, whilst the deformation introduced by the irregular parts renders unpredictability (Fig. 9).

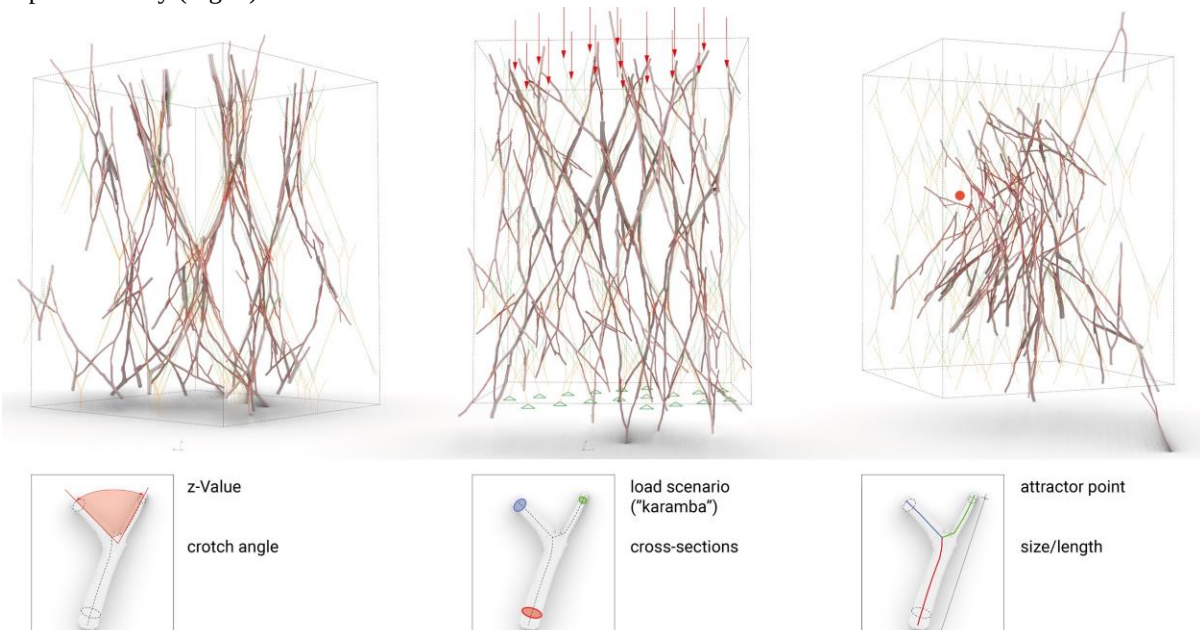


Fig. 9 different possibilities of sorting/replacing branches within the regular aggregation

Approximating the set more accurately by using multiple proto-parts in the initial aggregation produces an aggregated structure that is closer to the final result after relaxation, but limits the level of design control.

Currently the algorithm employed (Wasp) spontaneously chooses available proto-parts to aggregate following a rule-set of rotational angles. When applying the same rule set from the regular aggregation similar forms emerge without actually connecting open ends. As a result endlessly bifurcating strands occur. By tracking the indices of consecutive parts in the aggregation sequence, mapped against the indices of available connections (branch ends), it is possible to trace the formation of pseudo tetrahedral cells. This recording of topologies allows to add links to ends to be connected in order to form tetrahedral cells and exclude these from the available ends to connect to in the further aggregation process. Closing ends that would form the larger secondary cells, as they occur in the regular formation on the other hand, are mostly far apart. Forcing these to connect would result in large deformation in the relaxation process, counteracting the pre-approximation using proto-parts. Instead possible cross connections between open ends are found in the completed primary aggregation based on vicinity and adequate angle of the respective end segments (Fig. 10).

Those possible connections occur mainly at the periphery, leaving large areas within as unconnected strands.



Fig. 10 “Wasp” aggregations with same number of proto-parts but different relaxation processes

This design method allows to find an aggregation much closer to the final form after relaxation. The random selecting of parts reduces the level of control but resembles a higher continuity, closer to the logic of the grown parts. A higher number of proto-parts used increases global continuity, whereas fewer proto-parts allow more design control in the process of populating the actual parts (Fig 11.)

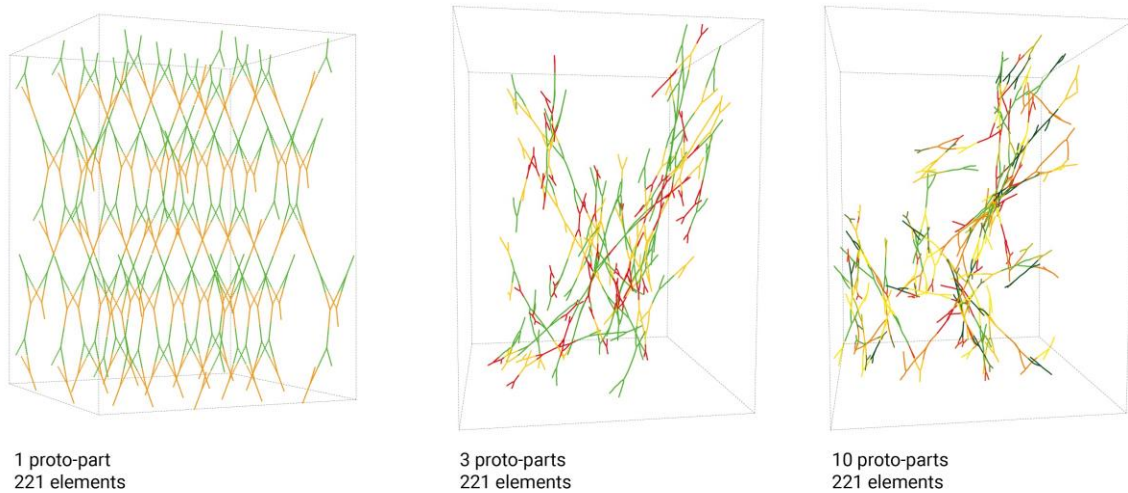


Fig.

11 “Wasp” aggregations with same count of elements but varying number of proto-parts

4. Conclusion and Outlook

The resulting methodology provides a flexible workflow for exploring formations based on branch elements. It offers a complete workflow linking geometry acquisition and rationalization, structure generation and further processing towards CNC fabrication (Fig. 12). Moreover, it offers a framework for exploring the threshold between regularity and irregularity, by creating a process where the amount of repetition and differentiation of the different elements can be edited by the user, hence allowing different levels of accuracy according to the requirements of different phases.

Simultaneously improvements of the aggregation algorithm (Wasp) and the digital workflow are being developed. Aggregation principles will be introduced according to which goals can be set to control the selection of parts in the iterative aggregation process. Goals include:

- Attempting to close loops - find best fitting part to connect to open ends based on geometric criteria to achieve maximum continuity and minimum cell size.
- Choose parts to align with global force patterns
- Field aggregation with attractor / repulsor conditions to control density

The endlessly bifurcating strands emerging from spontaneous aggregation of proto parts offer the potential to introduce cross-connections between members passing one another in close vicinity. These complementary joints introduce an alternative / additional method for bracing. This significant alteration to the initial design intent is a result of findings in the experimentation process.

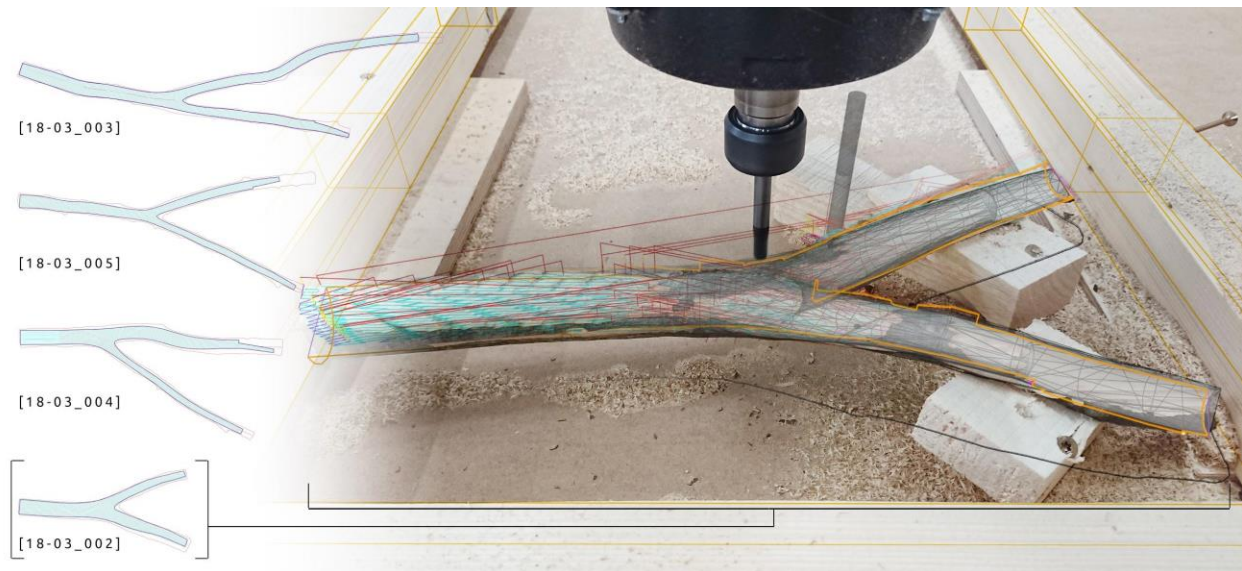


Fig. 12 3-Axis-CNC-milling of a Y-shaped branch fork

At the moment of writing this paper, work is being directed towards the production of a robotically-fabricated full-scale demonstrator. The framework presented in the paper is being further extended to implement required information for robotic milling of joints.

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